# Physics 111B: Brownian Motion

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**Abstract**

In this experiment, we investigated Brownian motion and intracellular transport to connect microscopic random motion with fundamental principles of statistical mechanics and biophysics. In the first part, we replicated Jean Perrin's classic confirmation of Einstein's theory of Brownian motion by tracking the trajectories of polystyrene microspheres suspended in fluids of varying viscosity and molecular composition. Using video microscopy and automated particle tracking, we extracted particle displacements and computed diffusion coefficients from mean-squared displacement analyses, examining their dependence on particle size and fluid viscosity. These results were compared with theoretical predictions from the Einstein–Langevin framework to evaluate the validity of the overdamped, inertia-less approximation and to identify experimental sources of deviation. In the second part, we studied intracellular motion in living onion epidermal cells, distinguishing passive Brownian diffusion from active, motor-driven transport. Vesicle trajectories were analyzed to characterize velocity distributions, directionality, and transport mechanisms associated with actin–myosin dynamics. Together, these investigations demonstrate how Brownian motion underlies both equilibrium diffusion and nonequilibrium biological transport, highlighting the power of quantitative microscopy and statistical analysis in probing physical processes at the cellular scale.

# 1 Introduction

Brownian motion is the continuous, random motion of small particles suspended in a fluid, caused by thermal collisions with the surrounding molecules. Although first systematically observed by Robert Brown in 1827, the phenomenon lacked a physical explanation until the development of molecular kinetic theory in the early twentieth century. In 1905, Albert Einstein and, independently, Marian Smoluchowski provided a quantitative description of Brownian motion, demonstrating that the observed fluctuations arise naturally from molecular bombardment. Einstein's work established a direct connection between microscopic particle trajectories and macroscopic thermodynamic quantities, providing one of the earliest experimentally testable predictions of atomic theory. Jean Perrin's subsequent experiments confirmed these predictions and played a crucial role in establishing the physical reality of atoms.

In Einstein's framework, the motion of a spherical particle of radius immersed in a Newtonian fluid of viscosity $\eta$ at temperature T is characterized by a diffusion coefficient $D = \frac{k_B T}{6\pi\eta}$ which is the Stokes-Einstein relation. This comes from Einstein's diffusion theory with Stokes' law for viscous drag and assumes motion in the overdamped, low Reynolds number regime, where inertial effects are negligible. In the first part of this experiment, we replicate Perrin's classic measurements using modern optical microscopy and automated particle tracking. Polystrene microscopes of varying diameters are suspended in fluids with different viscosoies and solute moleculars weights, which allow systematic tests of the Stokes-Einstein relation. Be measuring mean-squared displacements and comparing the resulting diffusion coefficients to theoretical predictions, we can evaulate the validity of the overdamped approximation and identify sources of experimental deviation.

In the second part of the experiment, we extend these concepts to a biological system by examining intracellular motion in living onion epidermal cells. Within the cytoplasm, some vesicles undergo passive Brownian diffusion, while other exhibit directed motion driven by molecular motors such as myosin moving along actin filaments. The coexistence of diffusive and active transport provides a natural contrast between equilibrium Browninan motion and nonequilibrium, energy-consuming processes. Analyzing vesicle trajectories and velocities enables direct comparison between thermal diffusion and

motor-driven transport within the complex cellular enviroment.
Overall, this experiment connects foundational results from statmech, including the Stokes-Einstein and Langevin descriptions of Brownian motion to contemporary biophysical phenomena.

# 2 Theory

Micron-scale particles suspended in a liquid undergo continual, irregular motion due to random, thermal collisions with surrounding solvent molecules. This phenomenon, Brownian motion, is well described statistically and provides a direct link between microscopic thermal agitation and macroscopic transport parameters such as the diffusion coefficient.

## 2.1 Overdamped dynamics, Stokes drag, and the Langevin picture

the regime relevant to this lab (micron-sized beads in water-like fluids), viscous forces dominate inertial forces. For a spherical particle of radius $a$ moving at velocity $v(t)$ throught a Newtonian fluid of viscosity $\eta$, Stokes' law gives the drag force as:

$$F_{drag} = -\gamma v, \gamma = 6\pi\eta a \tag{1}$$

A standard model for Brownian motion is the Langevin equation, which decomposes the force into deterministic drag plus a random thermal force $\xi(t)$"

$$m\frac{dv}{dt} = -\gamma v(t) + \xi(t), \langle \xi(t) \rangle = 0 \tag{2}$$

The inertial relaxation time is

$$\tau = \frac{m}{\gamma} \tag{3}$$

When $\Delta t \gg \tau$, the exponential inertial transient term in the general Langevin solution is negligible, and the motion appears purely diffusive at the sampling times used in video microscopy. This is the "inertia-less" (overdamped) limit emphasized in the lab manual.

## 2.2 Diffusion and mean-squared displacement

In the diffusive limit, particle position $r(t)$ satisfies the hallmark linear MSD scaling. The lab manual highlights the core relationship as:

$$\left\langle |\mathbf{r}(t+\Delta t) - \mathbf{r}(t)|^2 \right\rangle = 2D\,\Delta t. \tag{4}$$

More generally, in $d$ spatial dimensions the diffusion prediction is

$$\left\langle |\mathbf{r}(t+\Delta t) - \mathbf{r}(t)|^2 \right\rangle = 2dD\,\Delta t, \tag{5}$$

so for tracked microscope data in 2D (measured $x, y$), one expects

$$\left\langle \Delta x^2 + \Delta y^2 \right\rangle = 4D\,\Delta t, \tag{6}$$

$$\left\langle \Delta x^2 \right\rangle = \left\langle \Delta y^2 \right\rangle = 2D\,\Delta t. \tag{7}$$

This linear-in-$\Delta t$ behavior is the primary method used in Investigation I to estimate $D$ from trajectories by fitting the mean-squared displacement (MSD) versus $\Delta t$.

### 2.3 Stokes-Einstein Relation

Einstein connected diffusion to thermal energy and viscous dissipation. Combining his result with Stokes drag yields the Stokes–Einstein relation for a spherical particle:

$$D = \frac{k_B T}{6\pi\eta a} \tag{8}$$

This predicts the key dependencies explored in the lab: (i) increasing viscosity $\eta$ decreases $D$, and (ii) increasing particle radius $a$ decreases $D$, at fixed temperature $T$. Comparing measured $D$ values across bead sizes and fluids is therefore a direct test of Einstein's model under real experimental conditions.

### 2.4 Gaussian Displacement Statistics

The manual describes an alternative estimate of $D$ based on the distribution of displacements over a fixed interval. For ideal Brownian motion, each Cartesian displacement is Gaussian:

$$P(\Delta x; \Delta t) = \frac{1}{\sqrt{4\pi D\Delta t}} \exp\left(-\frac{(\Delta x)^2}{4D\Delta t}\right), \qquad \left\langle (\Delta x)^2 \right\rangle = 2D\Delta t. \tag{9}$$

In 2D, the joint displacement distribution factorizes,

$$P(\Delta x, \Delta y; \Delta t) = \frac{1}{4\pi D\Delta t} \exp\left(-\frac{(\Delta x)^2 + (\Delta y)^2}{4D\Delta t}\right). \tag{10}$$

## 3 Experimental Procedure

### 3.1 Part 1: Brownian motion of microbeads

Brownian motion measurements were carried out on an inverted light microscope coupled to a CCD camera and particle-tracking software, using the 20× objective for all datasets. Polystyrene microspheres with known diameters were dispersed in liquids whose viscosities were set by preparing glycerol–water and PVP solutions at several concentrations. For each condition, a small aliquot of the bead suspension was pipetted onto a clean glass slide and sealed with a coverslip to create a wet mount. The sample was imaged under dark-field illumination to maximize bead contrast against the background, and the condenser/illumination settings were tuned so that static dust or surface features were not misidentified as moving beads. Before collecting data, the image scale was calibrated with an object of known dimension to convert pixel coordinates into micrometers; this calibration factor for the 20× objective was then entered into the tracking software. Time series videos were recorded while tracking multiple beads simultaneously, and the software output the bead trajectories as $(x(t), y(t))$ position data. These trajectories were subsequently used to compute mean-squared displacements and extract diffusion coefficients for each viscosity–bead size condition.

### 3.2 Part 2: Intracellular transport in onion cells

Intracellular transport measurements were performed using onion epidermal tissue as a simple living-cell model. A single, thin epidermal sheet was gently peeled from the inner face of an onion scale, laid flat in mounting medium on a microscope slide, and sealed with a coverslip. Imaging was carried out with the 20× objective in bright-field to clearly visualize vesicles/granules over time, and time-lapse videos were collected long enough to capture sustained directed runs. To quantify active transport, we generated kymographs by extracting intensity along a chosen transport "track," producing a space–time plot in which directed motion appears as straight streaks; the slope of each streak gives displacement per unit time and therefore the particle velocity. Pixel distances were converted to physical length using the microscope calibration, and the time axis was set by the camera frame interval, allowing slopes to be reported as velocities in $\mu m/s$

# 4 Data Analysis

## 4.1 Particle Diffusion

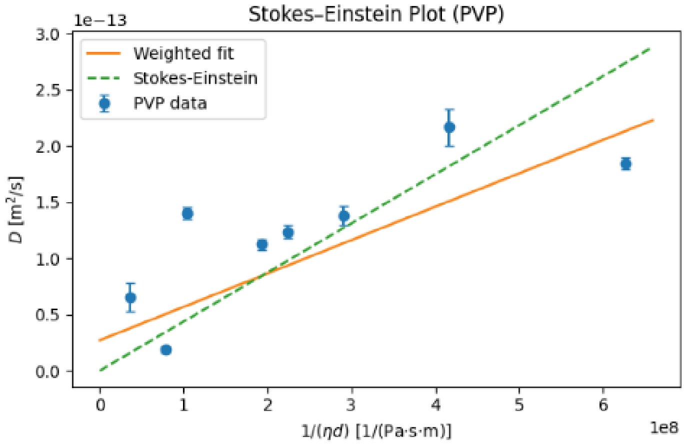


Figure 1: Figure 1: Stokes-Einstein Plot (PVP)

Tracked beads were computed using MSD, or equation (5) from earlier. Using our different viscosities $\eta$ and bead diameter $d$ from each test of our test cases, and from each tracker output file read the per-step time increment $Deltat$ and squared displacement $\Delta r^2$ we were able to compute the stepwise diffusion estimates using 2D brownian motion, $D_i = \Delta r_i^2 / 4\Delta t_i$, For each condition we then plotted the mean with error bars given by the standard error . The x-axis was constructed as $x = 1/\eta d$, motivated by the Stokes-Einstein relation from equation 8, so the theoretical prediction is a straight line:

$$D_{SE}(x) = \frac{k_B T}{3\pi} x \tag{11}$$

Finally, we overlaid a weighted linear fit obtained by minimizing $\chi^2 = \sum_j \frac{(D_j - (m x_j + b))^2}{\sigma_{D,j}^2}$ to compare scaling against the Stokes-Einstein expectation.

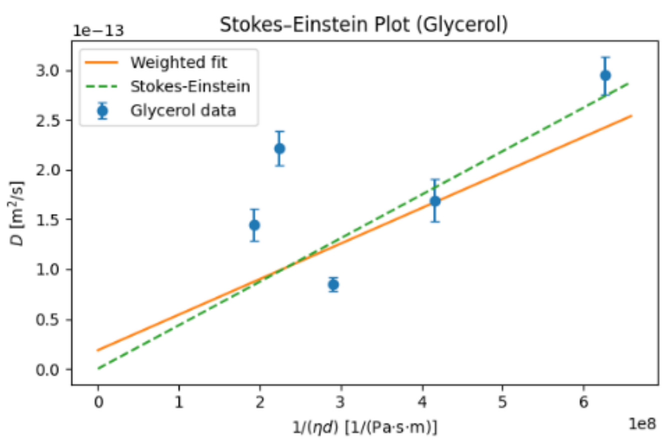


Figure 2: Figure 2: Stokes-Einstein Plot (Glycerol)

Tracked bead diffusion coefficients were computed from the mean-squared displacement (MSD), using Eq. (5) from earlier for 2D Brownian motion,

$$\langle \Delta r^2 \rangle = 4D \, \Delta t.$$

For each glycerol test case (each viscosity $\eta$ and bead diameter $d$ condition), the particle tracker output provided the per-step time increment $\Delta t_i$ and squared displacement $\Delta r_i^2$. We therefore formed stepwise diffusion estimates

$$D_i = \frac{\Delta r_i^2}{4\Delta t_i},$$

and for each condition computed the average diffusion coefficient

$$\bar{D} = \frac{1}{N}\sum_{i=1}^{N} D_i,$$

with uncertainty given by the standard error of the mean,

$$\sigma_{\bar{D}} = \frac{s_D}{\sqrt{N}}, \qquad s_D = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(D_i - \bar{D})^2}.$$

To test Stokes–Einstein scaling, we plotted $\bar{D}$ versus

$$x = \frac{1}{\eta d},$$

which is motivated by the Stokes–Einstein relation (Eq. (8)),

$$D = \frac{k_B T}{6\pi\eta a} = \frac{k_B T}{3\pi\eta d} \quad (a = d/2),$$

so the theoretical prediction is a straight line

$$D_{\mathrm{SE}}(x) = \left(\frac{k_B T}{3\pi}\right) x.$$

Finally, we overlaid a weighted linear fit $D_{\mathrm{fit}}(x) = mx + b$ to the measured points (using weights $w_j = 1/\sigma^2_{\bar{D},j}$) to compare the experimental scaling against the Stokes–Einstein expectation.

### 4.2 Analysis of Graphs

When comparing solvents directly, the glycerol data show larger scatter and stronger differences from the theoretical line for some conditions than the PVP data. This may reflect the sensitivity of the measurement to experimental conditions (focus height, wall proximity, drift) and, potentially, differences in how the bead "feels" the local fluid environment: PVP is a polymer solution that can introduce microstructural or viscoelastic effects at short times, whereas glycerol is a small-molecule Newtonian fluid that more closely matches the simplest Stokes–Einstein assumptions. In practice, both datasets still capture the correct qualitative scaling, but neither perfectly matches the ideal prediction, suggesting that experimental systematics plus possible solvent-dependent microscopic effects limit agreement more than any breakdown of the basic diffusion model.

### 4.3 Diffusion Coefficient Tables and Percent Error

Tables 1 and 2 compare the experimentally measured diffusion coefficients $D_{\mathrm{exp}}$ to theoretical predictions $D_{\mathrm{th}}$ for beads in glycerol and PVP solutions, respectively. All diffusion coefficients are reported in units of $10^{-13}$ m$^2$/s. The experimental values $D_{\mathrm{exp}}$ were obtained from tracked bead motion using the 2D Brownian-motion MSD relation,

$$\langle \Delta r^2 \rangle = 4D\,\Delta t,$$

by computing stepwise estimates $D_i = \Delta r_i^2/(4\Delta t_i)$ and averaging to get $\bar{D}$, with uncertainty reported as the standard error of the mean. The theoretical values were computed using the Stokes–Einstein relation written in terms of bead diameter $d$ (with $a = d/2$),

$$D_{\mathrm{th}} = \frac{k_B T}{6\pi\eta a} = \frac{k_B T}{3\pi\eta d},$$

assuming constant room temperature $T$. Percent error is reported as a signed deviation,

$$\% \text{ error} = 100 \, \frac{D_{\text{exp}} - D_{\text{th}}}{D_{\text{th}}},$$

so negative values indicate $D_{\text{exp}} < D_{\text{th}}$ and positive values indicate $D_{\text{exp}} > D_{\text{th}}$.

In glycerol (Table 1), two conditions show close agreement with Stokes–Einstein (errors 10%), while several conditions deviate substantially, indicating that systematic effects (e.g., wall hindrance, residual drift/flow, or tracking bias) likely influenced the measured displacements for those datasets. In PVP (Table 2), one condition is close to theory, but multiple conditions show large positive deviations, consistent with a combination of experimental systematics and the possibility that polymer microstructure/nonequilibrium effects altered the effective drag at the bead scale.

Table 1: Experimental and theoretical diffusion coefficients for beads in glycerol solutions. Diffusion coefficients are reported in units of $10^{-13}$ m$^2$/s.

| $\eta$ (mPa·s) | $d$ ($\mu$m) | $D_{\text{exp}}$ ($\times 10^{-13}$) | $D_{\text{th}}$ ($\times 10^{-13}$) | % error |
|---|---|---|---|---|
| 2.50 | 0.96 | $1.692 \pm 0.216$ | 1.820 | −7.0% |
| 4.65 | 0.96 | $2.216 \pm 0.169$ | 0.978 | 126.6% |
| 0.96 | 1.66 | $2.945 \pm 0.189$ | 2.741 | 7.4% |
| 1.66 | 2.07 | $0.845 \pm 0.069$ | 1.271 | −33.5% |
| 2.50 | 2.07 | $1.448 \pm 0.164$ | 0.844 | 71.6% |

Table 2: Experimental and theoretical diffusion coefficients for beads in PVP solutions. Diffusion coefficients are reported in units of $10^{-13}$ m$^2$/s.

| $\eta$ (mPa·s) | $d$ ($\mu$m) | $D_{\text{exp}}$ ($\times 10^{-13}$) | $D_{\text{th}}$ ($\times 10^{-13}$) | % error |
|---|---|---|---|---|
| 1.66 | 0.96 | $1.844 \pm 0.057$ | 2.741 | −32.7% |
| 2.50 | 0.96 | $2.171 \pm 0.163$ | 1.820 | 19.3% |
| 4.65 | 0.96 | $1.238 \pm 0.056$ | 0.978 | 26.6% |
| 13.20 | 0.96 | $0.186 \pm 0.026$ | 0.345 | −46.1% |
| 1.66 | 2.07 | $1.380 \pm 0.080$ | 1.271 | 8.6% |
| 2.50 | 2.07 | $1.123 \pm 0.051$ | 0.844 | 33.1% |
| 4.65 | 2.07 | $1.402 \pm 0.053$ | 0.454 | 208.8% |
| 13.20 | 2.07 | $0.653 \pm 0.126$ | 0.160 | 308.1% |

## 4.4 Molecular Motors Analysis

### 4.4.1 Kymographs

Figure 3: Figure 2: Kymographs

### 4.5 Molecular motors (onion epidermal cells): analysis with measured kymograph velocity

**Velocity extraction from the kymograph.** With spatial calibration $\ell_{\text{pix}} = 0.24\ \mu$m/pixel and frame interval $\Delta t_{\text{frame}} = 1.125$ s/frame, the velocity from a traced line segment on a kymograph is

$$v = \frac{\Delta x}{\Delta t} = \frac{\Delta x_{\text{pix}}\ell_{\text{pix}}}{\Delta t_{\text{pix}}\Delta t_{\text{frame}}}.$$

For the measured directed event $v_1 = 16.8\ \mu$m/s over $\Delta t = 1.125$ s,

$$\Delta x = v_1 \Delta t = (16.8)(1.125) = 18.9\ \mu\text{m}, \qquad \Delta x_{\text{pix}} = \frac{\Delta x}{\ell_{\text{pix}}} = \frac{18.9}{0.24} \approx 79 \text{ pixels}.$$

**Average velocity and comparison to Brownian motion.** Taking the directed event as representative, the average active-transport speed is

$$\bar{v}_{\text{active}} \approx 16.8\ \mu\text{m/s}.$$

A diffusion-based comparison can be made using the RMS frame-to-frame speed for Brownian motion:

$$v_{\text{rms}} \approx \frac{\sqrt{\langle \Delta r^2 \rangle}}{\Delta t} = \frac{\sqrt{4D\Delta t}}{\Delta t} = 2\sqrt{\frac{D}{\Delta t}}.$$

Using $\Delta t = 1.125$ s and a typical bead diffusion coefficient from our Brownian runs $D \sim (0.8\text{–}3.0) \times 10^{-13}$ m$^2$/s $= (0.08\text{–}0.30)\ \mu$m$^2$/s,

$$v_{\text{rms}} \sim 2\sqrt{\frac{0.08\text{–}0.30}{1.125}} \approx 0.53\text{–}1.03\ \mu\text{m/s}.$$

Therefore, the directed intracellular motion ($\sim 16.8\ \mu$m/s) is roughly

$$\frac{\bar{v}_{\text{active}}}{v_{\text{rms}}} \sim 16\text{–}32$$

times faster than diffusion-driven frame-to-frame motion, and is also strongly directional (persistent), unlike Brownian fluctuations.

**Work to transport a vesicle (drag-limited estimate).** Model the vesicle as a sphere of radius $a$ moving at speed $v$ through cytosol of viscosity $\eta_{\text{cyt}}$. Stokes drag gives

$$F_{\text{drag}} = 6\pi\eta_{\text{cyt}} a v.$$

Assuming approximately constant $v$ along a path length $L$, the mechanical work against viscous drag is

$$W \approx F_{\text{drag}} L = 6\pi\eta_{\text{cyt}} a v L.$$

Using the measured segment length $L = \Delta x = 18.9\ \mu$m and $v = v_1 = 16.8\ \mu$m/s,

$$W_{\text{seg}} \approx 6\pi\eta_{\text{cyt}} a \,(16.8 \times 10^{-6})\,(18.9 \times 10^{-6})\ \text{J}.$$

(For perimeter-to-center transport, replace $L$ by the full transport distance $L_{\text{cell}}$.)

**ATP hydrolysis energy and comparison.** Let the free energy available per ATP under cellular conditions be $\Delta G_{\text{ATP}}$ (J/mol). Per molecule,

$$\epsilon_{\text{ATP}} = \frac{|\Delta G_{\text{ATP}}|}{N_A}.$$

The number of ATP equivalents required to supply the mechanical work is then

$$N_{\text{ATP}} \frac{W}{\varepsilon\, \epsilon_{\text{ATP}}} = \frac{6\pi\eta_{\text{cyt}} a v L}{\varepsilon} \cdot \frac{N_A}{|\Delta G_{\text{ATP}}|},$$

where $\varepsilon \in (0, 1)$ is the mechanical efficiency of the motor system.

**Minimum number of motors required (force bound).** A conservative lower bound on the number of motors simultaneously engaged is obtained by comparing drag force to the maximum force per motor $F_{\max}$:

$$N_{\min}^{(F)} = \left\lceil \frac{F_{\text{drag}}}{F_{\max}} \right\rceil = \left\lceil \frac{6\pi\eta_{\text{cyt}}av}{F_{\max}} \right\rceil.$$

Using $v = v_1$,

$$N_{\min}^{(F)} = \left\lceil \frac{6\pi\eta_{\text{cyt}}a(16.8 \times 10^{-6})}{F_{\max}} \right\rceil.$$

This bound increases if $\eta_{\text{cyt}}$ or $a$ are larger, and decreases if the motion is partly due to bulk cytoplasmic streaming rather than a single vesicle being actively hauled.

# 5 Systematic Uncertainties and Errors

This experiment compares measured diffusion coefficients and intracellular transport speeds against simple physical models (Stokes–Einstein diffusion and Stokes drag). The dominant uncertainties are systematic rather than purely statistical, because both particle tracking and microscopy introduce biases that can shift extracted parameters in a condition-dependent way.

## 5.1 Systematic errors in Brownian-bead diffusion measurements

**(1) Localization error (static tracking noise).** Finite position precision adds a constant offset to the measured MSD,

$$\text{MSD}_{\text{obs}}(\Delta t) = 4D\,\Delta t + 4\sigma^2,$$

so naive slope estimates using short time lags can be biased, and using stepwise estimates $D_i = \Delta r_i^2/(4\Delta t_i)$ can be inflated when $4\sigma^2$ is comparable to $4D\Delta t$.

**(2) Motion blur (finite camera exposure).** During the exposure time, particles move while being imaged; this "dynamic error" can alter the apparent step distribution and bias diffusion estimates, especially for faster diffusion or longer exposures.

**(3) Drift or residual bulk flow.** Stage drift, convection, or slow flow in the wet mount adds directed motion on top of diffusion. In that case,

$$\text{MSD}(\Delta t) = 4D\,\Delta t + (v\,\Delta t)^2,$$

so even a small drift speed $v$ can strongly inflate the inferred $D$ at longer lags. This is a leading explanation for very large *positive* percent errors (measured $D_{\text{exp}} \gg D_{\text{th}}$).

**(4) Near-wall hindered diffusion (confinement/hydrodynamic wall effects).** Beads often move close to the slide or coverslip, increasing hydrodynamic drag and suppressing diffusion relative to bulk Stokes–Einstein. This produces *negative* deviations ($D_{\text{exp}} < D_{\text{th}}$) and depends strongly on bead height above the surface.

**(5) Viscosity and temperature mismatch.** The theoretical prediction

$$D_{\text{th}} = \frac{k_B T}{3\pi\eta d}$$

depends directly on the *actual* viscosity $\eta$ of the sample (which is sensitive to concentration and temperature). Any uncertainty in $\eta$ propagates linearly into $D_{\text{th}}$. In addition, polymer solutions (PVP) can exhibit scale- or time-dependent effective drag, so the bead may not experience the nominal bulk viscosity at the timescales used for analysis.

**(6) Calibration and bead-size uncertainty.** A fractional error in the pixel-to-length calibration $\ell_{\rm pix}$ enters the inferred diffusion coefficient approximately as

$$\frac{\delta D}{D} \approx 2\frac{\delta \ell_{\rm pix}}{\ell_{\rm pix}},$$

because displacements enter as squared distances. Any uncertainty in bead diameter $d$ also propagates directly to $D_{\rm th} \propto 1/d$.

## 5.2 Systematic errors in intracellular transport (kymograph velocities)

Velocities were extracted from the slope of directed tracks on kymographs using

$$v = \frac{\Delta x_{\rm pix} \ell_{\rm pix}}{\Delta t_{\rm pix} \Delta t_{\rm frame}}.$$

For a representative event, $v_1 = 16.8$ $\mu$m/s corresponds to $\Delta x = 18.9$ $\mu$m over $\Delta t = 1.125$ s (about 79 pixels in $x$ for $\ell_{\rm pix} = 0.24$ $\mu$m/pixel).

**(1) Manual line placement and path selection.** If trajectories are not perfectly straight (accelerations, pauses, switching tracks), fitting a single straight line introduces bias. The largest systematic uncertainty is often the subjective choice of which segment to fit and where the track begins/ends.

**(2) Projection and out-of-plane motion.** Kymographs measure motion projected onto the chosen line in the image plane. Any 3D motion or motion not aligned with the drawn line leads to an underestimate of the true speed, while apparent motion can also be generated by focus changes or organelles moving in/out of the focal plane.

**(3) Background cytoplasmic streaming and global drift.** Bulk cytoplasmic flow can advect organelles, producing slopes on kymographs even without single-vesicle motor hauling. If streaming is present, the measured $v$ may represent a superposition of motor-driven transport and advective flow.

**(4) Timing and calibration uncertainties.** Errors in $\ell_{\rm pix}$ and $\Delta t_{\rm frame}$ propagate directly:

$$\frac{\delta v}{v} \approx \sqrt{\left(\frac{\delta \ell_{\rm pix}}{\ell_{\rm pix}}\right)^2 + \left(\frac{\delta \Delta t_{\rm frame}}{\Delta t_{\rm frame}}\right)^2 + \left(\frac{\delta(\Delta x_{\rm pix}/\Delta t_{\rm pix})}{\Delta x_{\rm pix}/\Delta t_{\rm pix}}\right)^2}.$$

If $\Delta x_{\rm pix}$ and $\Delta t_{\rm pix}$ are read manually, pixel-level rounding can be non-negligible for short segments.

## 5.3 Systematics in derived motor energetics (work, ATP, motor count)

When estimating work against viscous drag using Stokes drag,

$$F_{\rm drag} = 6\pi\eta_{\rm cyt} a v, \qquad W \approx F_{\rm drag} L = 6\pi\eta_{\rm cyt} a v L,$$

the largest systematic uncertainties are (i) the effective cytosol viscosity $\eta_{\rm cyt}$, (ii) the vesicle size $a$, (iii) whether the motion is purely drag-limited (vs. elastic/active remodeling), and (iv) whether the measured $v$ includes background flow (streaming). Because $W$ scales linearly with each factor, the relative uncertainty is approximately

$$\frac{\delta W}{W} \approx \sqrt{\left(\frac{\delta \eta_{\rm cyt}}{\eta_{\rm cyt}}\right)^2 + \left(\frac{\delta a}{a}\right)^2 + \left(\frac{\delta v}{v}\right)^2 + \left(\frac{\delta L}{L}\right)^2}.$$

Comparisons to ATP hydrolysis energy and estimates of the minimum number of motors therefore inherit these same systematics, in addition to uncertainty in the available free energy per ATP and the effective mechanical efficiency of the motor system.